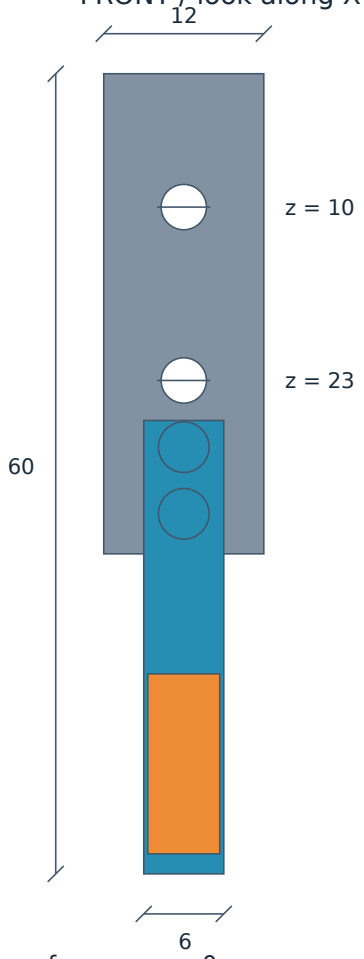
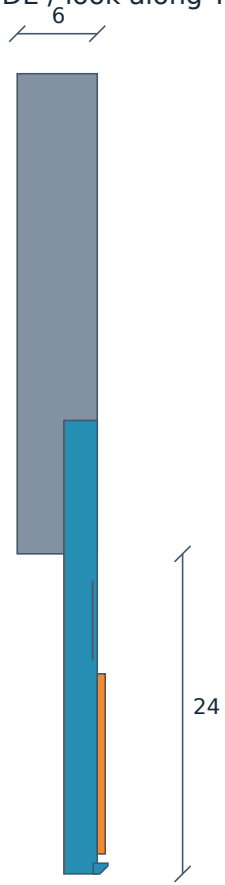
01 / Assembly and mounting datums

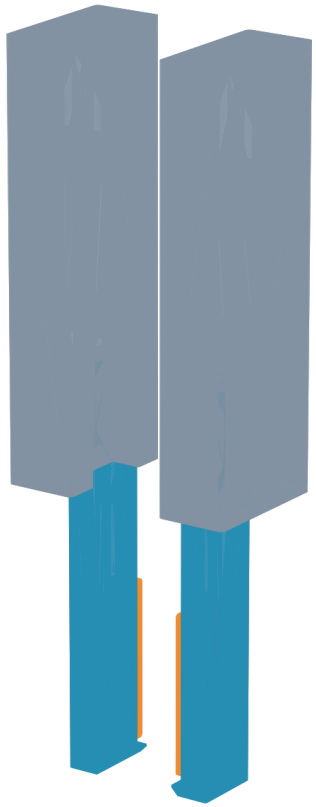
MHZL2-16D2 replacement fingertip concept | REV A | 2026-11-01

FRONT / look along X



SIDE / look along Y





Base reference: z = 0

Mount hole centers: y = 0; z = 10, 23

2X DIA 3.4 through X (ASSUMED)

Hole pitch = 13; carrier section = 12 x 6

Insert screws: z = 28, 33; y = 0

x = 0: carrier inner face

Insert body x = -2.5 to 0

Working zone z = 36 to 60

Toe ledge z = 59.2

Toe distal end z = 60

CLOSED ASSEMBLY / reference placement

Pad gap 7.0 closed / 19.0 open (uncompressed)

Toe gap 6.6 closed / 18.6 open

Required inner mounting-face gap: 8.2 closed

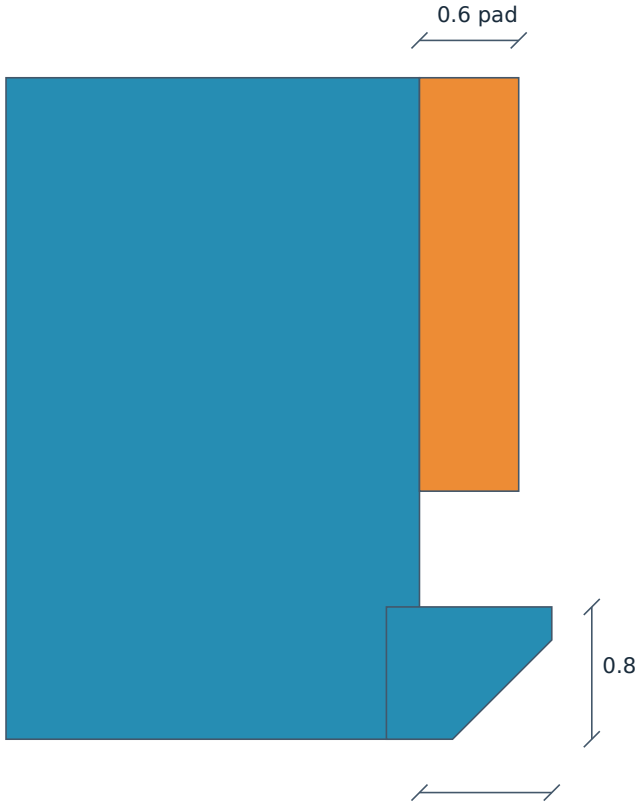
Travel: 6.0 per jaw; screw hardware not modeled

02 / Insert details and interface control

MHZL2-16D2 replacement fingertip concept | REV A | 2026-11-01

TOE / enlarged side section

Orange: compliant pad; blue: rigid insert



0.6 x 0.6 distal lead-in chamfer

0.7 axial clearance: pad end to toe ledge

0.2 toe underlap beyond uncompressed pad

Insert outer longitudinal edges R0.2

LOCAL COORDINATES / same component used twice

Datum origin	(0, 0, 0): inner carrier face, width center, base
Axis convention	+X inward; +Y across width; +Z toward workpiece
Mount hole centers	(0, 0, 10), (0, 0, 23); axes parallel X
Insert envelope	x -2.5..0; y -3..3; z 26..60
Carrier pocket	x -2.5..0; y -3.1..3.1; z 25.9..36
Insert fastening	2X DIA 2.2 through; DIA 3.8 counterbore depth 1.2
Carrier fastening	2X DIA 1.6 pilot for M2 tap; z 28, 33
Pad envelope	x 0..0.6; y -2.7..2.7; z 45..58.5; R0.3 corners
Stud saddle	R2.5; depth 0.35; z 38..44; cylinder axis Z
Saddle axis	x = 2.15; y = 0; rigid face, above pad
Toe load ledge	z = 59.2; x 0..0.8; y -3..3
Assembly left	local translated X = -4.1 closed / -10.1 open
Assembly right	rotate 180 deg about Z, then X = +4.1 / +10.1

INTEGRATION HOLD POINTS

Confirm actual closed face spacing; 8.2 is a design requirement, not measured.

Confirm base datum, hole axis, diameter, screws and mounting-face orientation.

J6-to-finger rigid transform is unknown; no J6 placement is asserted.

M2 low-head screw envelope must fit the 3.8 x 1.2 recess; verify engagement.

Pad: nominal TPU/urethane 60-80A; adhesive thickness must be included in 0.6.

Prototype only: validate grip force, toe strength, reach and neighboring-part clearance.

Toe needs under-edge access; it cannot enter a perfectly flush, gap-free interface.